Sum and Difference Formulas

Section 1: Sine Sum and Difference Identities

1 State the formula for $\sin(A+B)$.

a. $\sin A\cos B+\cos A\sin B$

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A+\sin B$

2 Find the exact value of $\sin(75^\circ)$ using $75^\circ=45^\circ+30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}+1}{2}$

d. $\frac{\sqrt{2}+1}{2}$

3 Simplify $\sin(x+\pi)$.

a. $-\sin x$

b. $\sin x$

c. $-\cos x$

d. $\cos x$

Section 2: Cosine Sum and Difference Identities

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4 State the formula for $\cos(A - B)$.

a. $\cos A \cos B + \sin A \sin B$

b. $\cos A \cos B - \sin A \sin B$

c. $\sin A \sin B - \cos A \cos B$

d. $\cos A - \cos B$

5 Find the exact value of $\cos(15^\circ)$ using $15^\circ = 45^\circ - 30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{4}$

d. $\frac{\sqrt{2}+1}{4}$

6 Simplify $\cos(x - \pi/2)$.

a. $\sin x$

b. $-\sin x$

c. $\cos x$

d. $-\cos x$

Section 3: Tangent Sum and Difference Identities

7 State the formula for $\tan(A + B)$.

a. $\frac{\tan A + \tan B}{1 - \tan A \tan B}$

b. $\frac{\tan A - \tan B}{1 + \tan A \tan B}$

c. $\frac{\tan A + \tan B}{1 + \tan A \tan B}$

d. $\tan A + \tan B$

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8 If $\tan A = 1/2$ and $\tan B = 1/3$, find $\tan(A + B)$.

a. 1

b. 5/6

c. 5/7

d. 2/3

9 Evaluate $\frac{\tan(80^\circ) - \tan(20^\circ)}{1 + \tan(80^\circ)\tan(20^\circ)}$.

a. $\sqrt{3}$

b. $1/\sqrt{3}$

c. 1

d. 0

Section 4: Compound Angle Problems with Given Ratios

10 Given $\sin\alpha = 3/5$ (Quadrant I) and $\cos\beta = -5/13$ (Quadrant II), find $\sin(\alpha + \beta)$.

a. 33/65

b. $-33/65$

c. 63/65

d. $-63/65$

11 For the same angles ($\cos\alpha = 4/5$, $\sin\alpha = 3/5$, $\cos\beta = -5/13$, $\sin\beta = 12/13$), find $\cos(\alpha + \beta)$.

a. $-56/65$

b. 56/65

c. $-16/65$

d. 16/65

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12 In which quadrant does $\alpha + \beta$ lie in the previous problems?

a. Quadrant II

b. Quadrant I

c. Quadrant III

d. Quadrant IV

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